
Part 1: The Four Pillars of Computational Thinking

Task 1: Decomposition (Break it Down)

1. Collect member identification data (Name).
2. Identify the specific equipment being used.
3. Capture the duration of use in minutes.
4. Perform mathematical calculation for the Usage Score.
5. Format and display the final report to the gym manager.

Task 2: Pattern Recognition (Identify Trends)

Score is always calculated by multiplying the minutes used by 1.5.

Task 3: Abstraction (Filter the Essentials)

Essential Variables (Monitor)	Irrelevant Variables (Ignore)
Member Name	Member's Outfit Color
Equipment Type (e.g., Treadmill)	Member's Age
Total Minutes Spent	Time of Arrival

Task 4: Algorithm Design (Step-by-Step Logic)

1. Start the program.
 2. Display a welcome message to the GymPulse staff.
 3. Prompt the user to input the member's name.
 4. Prompt the user to input the equipment used.
 5. Prompt the user to input the total minutes spent.
 6. Calculate the Usage Score by multiplying minutes by 1.5.
 7. Print a summary report showing the name, equipment, and calculated score.
 8. End the program.
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Part 2: Behavioral Design (IPO and Flowchart)

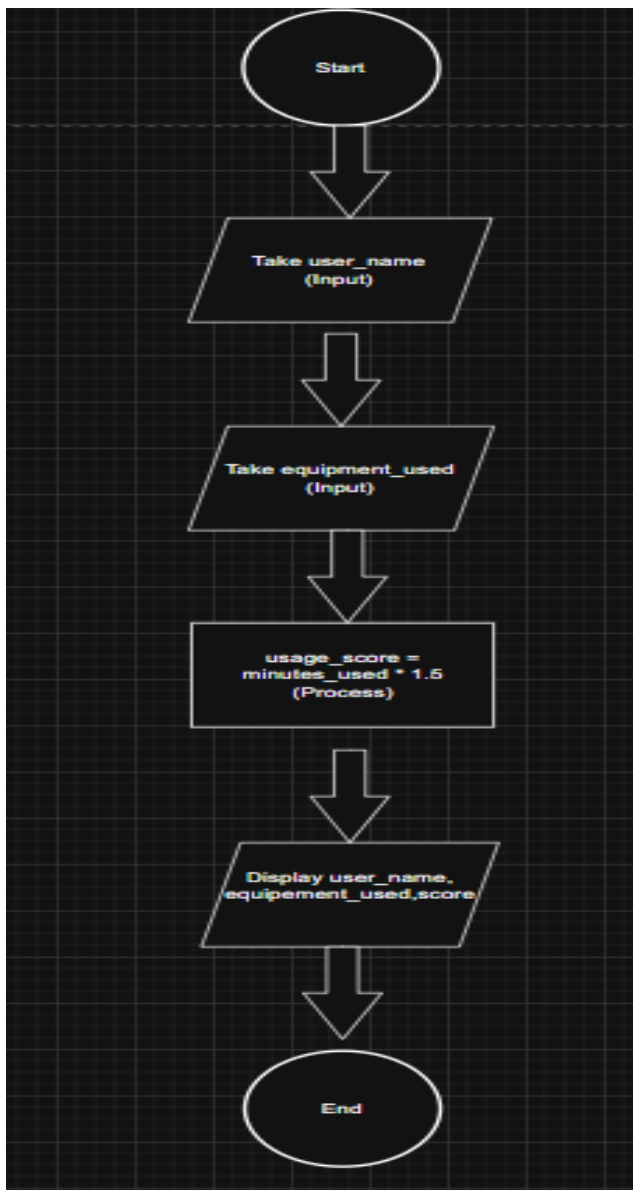
Task 2.1: The IPO Model

Input: Member_name, equipment_type, minutes_used

Process: $\text{usage_score} = \text{minutes_used} * 1.5$

Output: Summary display of member, machine, usage_score

Task 2.2: The Logic Flowchart Guide



Part 3: Implementation (Python Foundations)

```
# 1. INPUT: Get data from user
print("Welcome to GymPulse") # Fixed typo

# Get user's name
member_name = input("Enter member name: ")

# Get equipment used
equipment_used = input("Enter equipment used (e.g., Treadmill): ")

# Get minutes
minutes_used = int(input("Enter total minutes used: "))

# 2. PROCESS: Perform calculation
# The formula is always minutes * 1.5
usage_score = minutes_used * 1.5

# 3. OUTPUT: Display
print("\n" + "-" * 35)
print("          Calculated Score")
print("-" * 35)
print("Name:           ",member_name)
print("Equipment used:  ",equipment_used)
print("Minute Used:     ",minutes_used)
print("Usage Score:     ",format(usage_score, ".2f"))
print("-" * 35)
```